

# Food Microbiology

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## Science

May, 2026

**Short Title:** Food Microbiology  
**Faculty** Science and Computing  
**Department** Science  
**Cluster** Biology  
**Author** AHEARNE  
**Credits** 5

**Level:** Introductory

## Description of Module / Aims

This module will describe the role of microbes in food spoilage, food poisoning and food fermentations. Food preservation and quality assurance are also covered.

## Programmes

|           |                                                                             |           |
|-----------|-----------------------------------------------------------------------------|-----------|
| MBIO-0002 | BA (Hons) in Culinary Arts (WD_HCULA_B)                                     | 2 / 4 / M |
| BIOL-0024 | BSc (Hons) in Food Science and Innovation (WD_SFSIN_B)                      | 2 / 4 / M |
| MBIO-0002 | BSc (Hons) in Molecular Biology with Biopharmaceutical Science (WD_SMBIO_B) | 2 / 4 / M |
| MBIO-0002 | BSc in Food Science with Business (WD_SFDCS_D)                              | 2 / 4 / M |
| MBIO-0002 | BSc in Molecular Biology with Biopharmaceutical Science (WD_SMBIO_D)        | 2 / 4 / M |

## Indicative Content

- Relationships between micro-organisms and food
- Food poisoning and food-borne disease
- Food fermentations, probiotics, prebiotics
- Food spoilage
- Factors affecting growth of micro-organisms in food
- Methods of food preservation
- Food safety issues
- Microbiological testing of foods for general quality, indicator organisms and specific pathogens
- Review of major groups of micro-organisms important in food, e.g. Salmonella, E. coli O157 and other verotoxigenic E. coli, Campylobacter, S. aureus, Bacillus spp., Clostridium perfringens, Listeria spp., lactic acid bacteria, probiotics, enteric viruses
- Microbiological quality assurance methods; HACCP

## Learning Outcomes

*On successful completion of this module, a student will be able to:*

1. Describe the major groups of microbes as they relate to food.
2. Use standard laboratory techniques to examine food microbiologically for indicator organisms, specific pathogens and major groups of food related microbes.
3. Discuss food poisoning and its prevention.
4. Identify methods used for food preservation.
5. Use standard laboratory techniques to demonstrate food preservation methods.
6. Recognise different types of food fermentations and outline the role of starter cultures.
7. Discuss the potential of functional foods containing probiotics or prebiotics.
8. Record results of laboratory work using worksheets.

## Learning and Teaching Methods

- Lectures (incorporating active learning activities).
- Practical classes (incorporating lab-based assessment / project).

## Assessment Methods

|                           | Weighing | Outcomes Assessed |
|---------------------------|----------|-------------------|
| Final Written Examination | 50%      | 1,3,4,6,7         |
| Continuous Assessment     | 50%      |                   |
| <i>Practical</i>          | 50%      | 2,5,8             |

## Assessment Criteria

**<40%:** Little or no demonstration of an understanding of the fundamentals of food microbiology. Lack of competence with associated laboratory skills & practices & an inability to carry out critical thinking.

**40%-49%:** Will show a limited understanding of food microbiology. May have some difficulty with applying critical thinking and problem-solving. Will display basic ability with associated laboratory skills & practices.

**50%-59%:** As well as the above, the student will show a good understanding of the food microbiology covered with demonstration of some critical thinking & problem-solving skills. Will display good competence with associated laboratory skills & practices.

**60%-69%:** In addition to the above, will demonstrate more detailed knowledge of food microbiology, very good critical thinking skills & a high level of competence & efficiency in associated laboratory practicals.

**70%-100%:** All of the above to an excellent level.

## Learning Modes

| Learning Mode        | F/T Hours | P/T Hours |
|----------------------|-----------|-----------|
| Lecture              | 24        |           |
| Practical            | 24        |           |
| Independent Learning | 87        |           |

## Essential Material

- "Moodle ." [www.moodle.wit.ie](http://www.moodle.wit.ie). [www.moodle.wit.ie](http://www.moodle.wit.ie)
- Madigan, M.T., J.M. Martinko, D.A. Stahl, D. Clark, K.S. Bender and Buckley, D.H. \_Brock's Biology of Micro-organisms\_. 13th . Upper Saddle River, NJ: Prentice Hall / Pearson Education, 2011.

## Supplementary Material

- "Food Safety Authority of Ireland." [www.fsai.ie](http://www.fsai.ie). [www.fsai.ie](http://www.fsai.ie)
- Ray, B. and A. Bhuna. \_Fundamental Food Microbiology\_. London: Taylor & Francis, 2008.

## Requested Resources

- Lecture Room: Loose Seated
- Room Type: Biology Lab